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Research paper

Phantom elasticity reconstruction with Digital Image Elasto-Tomography

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ABSTRACT

Results from the application of a novel nonlinear hybrid reconstruction algorithm within a Digital Image Elasto-Tomography (DIET) system are presented. A hybrid reconstruction algorithm was optimized to solve for the elasticity distribution of two heterogeneous silicone phantoms using a shape-based parameterization. The hybrid algorithm achieved comparable performance to Combinatorial Optimization methods with significantly less computational expense. The specificity of three-parameter reconstruction was confirmed by successful reconstruction of a homogeneous silicone phantom, indicating the potential suitability of the DIET system for application to inclusion imaging in elastography.

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1. Introduction

Breast cancer is the most common form of cancer in the female population, with new cases in 2007 expected to represent a total of 26% of all female cancer diagnoses (Jemal et al., 2007). Although deaths from cancer of the lung and bronchus surpassed breast cancer fatalities among American women in 1987, breast cancer remains the second highest cause of female cancer death in the United States, with over 100 American women per day currently dying from the disease. Similar studies from New Zealand indicate that breast cancer is at least as significant an issue for women in countries outside the United States (New Zealand Health Information Service, 2002, 2006).

Breast self-examination is widespread for detecting the presence of suspicious lesions at an early stage. This

technique relies on the underlying mechanical properties of breast tissue, where tumors often present as a discrete mass significantly stiffer than the surrounding tissue (Chaplain et al., 2006). However, such an examination is only able to provide a qualitative assessment of tissue properties, limiting its usefulness for further diagnosis of tissue histology. In addition, a number of studies have challenged the effectiveness of breast self-examination as a useful tool for reducing the occurrence and fatality rates of advanced stage breast cancer (Newcomb et al., 1991; Holmberg et al., 1997; Thomas et al., 1997).

Despite the questionable effectiveness of manual palpation, the mechanical property contrast measured by breast self-examination has inspired investigation into means of quantification of soft tissue elastic properties. While experimentally tested soft tissue modulus values show large variations depending on a range of factors specific to each testing

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methodology, a clear distinction is apparent between healthy and cancerous breast tissue, with observed elastic contrast of over an order of magnitude (Sarvazyan, 1998; Krouskop et al., 1998; Wellman et al., 1999; Samani et al., 2007). These and other similar results have encouraged the development of a range of techniques that aim to determine the distribution of elastic properties within human tissue, known as elastography (Ophir et al., 1991). A number of these elastographic modalities are based on existing medical imaging techniques such as ultrasound or Magnetic Resonance Imaging (MRI), where a modified application of the hardware and/or software allows the generation of elasticity images that can assist in determining tissue pathology (Parker et al., 1990; Barr; Plewes et al., 2000; Van Houten et al., 2005).

Applications of stochastic techniques in the field of elastic property reconstruction are limited, although increasing in popularity as more powerful computer systems become mainstream. Genetic Algorithms (GAs) have been used in combination with finite element (FE) methods to solve for the mechanical properties and internal structure of composite materials (Matsuoka et al., 2001). Preliminary investigations have also been performed into the use of GAs for image reconstruction in Electrical Impedance Tomography (EIT) (Olmi et al., 2000). While such stochastic methods have been found to have significantly higher computational cost than more traditional Newton–Raphson methods, this increase is often offset by a clear improvement in reconstructed solution accuracy. Recent investigations into stochastic optimization in elastography have considered a theoretical approach to the application of a GA to soft tissue elastography (Zhang et al., 2006), while 2D studies have been performed using a combined GA and FE approach to solve for elasticity values in vascular soft tissue (Olmi et al., 2006).

The feasibility of elastographic property reconstruction from a set of boundary measurements has only been recently considered (Liu et al., 2005; Peters et al., 2005). These systems differ from ultrasound or MRI-based elastography techniques in the limited amount of data available for elastic property reconstruction. Promising results from simulated and basic phantom studies have indicated that elastographic reconstruction is possible using only a small subset of the large data sets used for full-volume based reconstruction techniques (Peters et al., 2006).

1.1. Digital Image Elasto-Tomography

Digital Image Elasto-Tomography (DIET) is a novel technique intended to detect areas of high tissue stiffness within the breast, where cancerous lesions can be identified due to their high elastic modulus values. Utilizing digital imaging sensors, image processing algorithms, and computer reconstruction techniques, DIET aims to generate a three-dimensional representation of elastic properties throughout the breast volume. The system is targeted at providing an inexpensive, comfortable screening test, where the associated hardware is compact enough to make portable screening units a reality. The three-dimensional nature of the resulting elastogram should allow accurate image registration between subsequent scans, assisting in tracking changes in internal breast structure over time.

This study considers the application of DIET reconstruction techniques to heterogeneous silicone phantoms of similar size to a female breast. Reconstruction algorithm refinement and performance are detailed, along with the first results from a surface motion based full-volume elastography system applied to tissue approximating phantoms. The overall goal is to create an initial computational and experimental proof of concept.

2. Methods

The three phantoms used in this study were cast from a platinum catalyzed soft silicone gel (Factor II Inc, AZ, USA). A ‘concentric’ heterogeneous phantom contained a 31 mm diameter hard cylinder embedded within a 70 mm high, 70 mm diameter soft cylinder, while a ‘spherical’ phantom contained a 40 mm diameter hard sphere embedded within a 70 mm high, 70 mm diameter cylinder. Finally, a homogeneous phantom consisted of a 70 mm high, 70 mm diameter cylinder. The mixtures for the soft ($E \approx 30$ kPa) and hard ($E \approx 100$ kPa) silicone were chosen to achieve a compromise between the stiffness values of actual breast tissue and stiffer, more durable materials conducive to experimental testing. All silicone phantoms had a single column of fifteen fiducial markers applied to the surface prior to motion imaging, colored to give the greatest possible contrast with the underlying silicone material. The three phantoms used were required to be axisymmetric due to limitations of the prototype experimental system, where only two cameras were available for motion imaging, thus limiting the region of effective data capture.

2.1. Experimental setup

The experimental actuation and image capture system performs two major functions, each controlled by separate computers. A schematic of the system is shown in Fig. 1. A ‘control’ computer is used for actuator control, motion feedback, and lighting tasks, while ‘imaging’ computers are used for motion capture from each of the cameras. A dSPACETM modular hardware control system (dSPACE GmbH, Germany) allows real-time control of the actuator and stroboscope.

The dSPACE system comprises a MATLABTM and SimulinkTM model incorporating specific dSPACE hardware control blocks. Once compiled, the control system code is downloaded from the control computer to the dSPACE hardware. An analog output port on the dSPACE unit provides a trigger signal to a stroboscope used for scene illumination. The dSPACE unit also provides frequency and amplitude control signals for the actuator. These signals are amplified to drive the electromagnets within the voice coil actuator. A velocity signal returned from the actuator provides the feedback path to complete a closed loop actuator control system. Control software and hardware is run in real time, allowing visualization of all input and output signals using the dSPACE Control Desk environment.

2.2. Actuation and motion capture

During the imaging experiment, phantoms were placed on a flat circular surface on top of the actuator that was configured

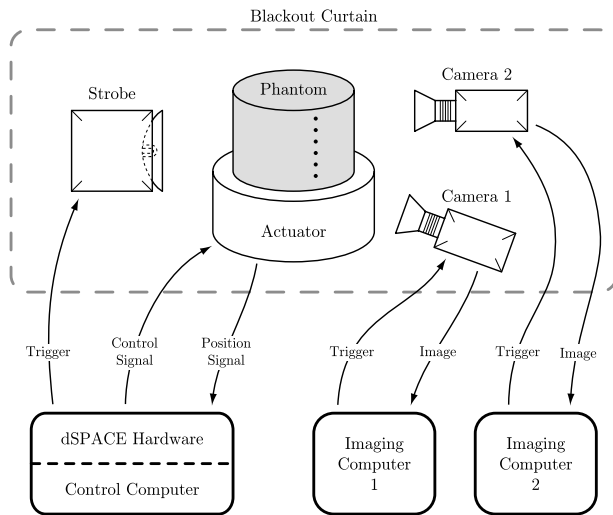


Fig. 1 – A schematic of the experimental setup for phantom motion capture.

to provide vertical sinusoidal motion. All phantoms were actuated at 100 Hz, with an amplitude of 0.5 mm. Both imaging cameras were set up with a field of view including the all fiducial markers on the phantom surface. The stroboscope was situated to give clear and even illumination of the markers from the perspective of both cameras. As any ambient light would upset the imaging process, the actuator, strobe and cameras were covered with a blackout curtain for the duration of the experiment.

Fig. 2 is a schematic showing how controlled strobe lighting is used to capture the full range of phantom motion, despite the high speed of actuation relative to the longer shutter speed and exposure time required to generate each digital image. Each row of the figure demonstrates the capture of an individual frame, with time, the horizontal coordinate. The sinusoidal line represents the steady state oscillation of a reference point on the phantom surface.

The time between strobe illumination is constant, and matched to the period of the actuator. This synchronization means that at each strobe offset, the reference point appears at a single, static location, allowing camera exposures longer than one motion cycle. After both cameras have downloaded their respective images, the phase offset between the strobe and actuation signals is adjusted.

After shifting the strobe phase, the phantom is illuminated at a different position in its motion cycle. This process allows the motion of any reference point to be tracked across the full range of its steady state motion. In this experiment, a strobe phase increment of 18° gives a total of 20 frames across one motion cycle. Fig. 3 shows 10 of the 20 captured surface motion frames from the concentric and spherical phantoms, where the heterogeneous phantom geometry and measured surface motion is detailed. The choice of time-harmonic actuation is made based on the fact that this method ensures that elastic strain energy is propagated throughout the imaging volume. As the image reconstruction process relies on the evidence provided by the response to this elastic strain energy, time-harmonic actuation thus ensures reasonable motion signal is achieved throughout the imaging volume, allowing accurate stiffness reconstruction.

2.3. Motion processing

Experimental motion capture results in sequential still images of all surface reference points across the complete phantom motion cycle. Because the DIET reconstruction approach is based on three-dimensional surface motion information, a conversion of reference point position from the two-dimensional image coordinate system into a three-dimensional, 'world space', coordinate system is required. Additionally, the measured experimental motion must be described in a manner that allows comparison with FE-simulated displacement fields.

2.3.1. Calibration and photogrammetry

Calibration of the camera system is performed prior to actuation and motion capture, using a specially developed calibration object. The mathematics used for this conversion are beyond the scope of this article, but are covered in detail in a number of texts, including those by Klette et al. (1998) and Trucco and Verri (1998). The output from the calibration process is a set of camera specific parameters that accurately describe the position and orientation of each camera, meaning the cameras cannot be moved between calibration and phantom imaging. The calibration matrices for each camera are used in conjunction with the captured frames to calculate a three-dimensional position in world space for every reference point over every frame i , termed $\bar{\mathbf{P}}^i$.

Fig. 4 demonstrates the result of the image to world-space conversion process, where the left and right images of a reference point motion track are shown in panels (a) and (b). The output from experimental motion capture is a vector for each reference point, $\bar{\mathbf{P}}$, where the world-space coordinates of the reference point are plotted in Fig. 4(c).

2.3.2. Reference point projection

It is not possible to achieve an exact match between the surface of a FE mesh and the surface of an experimental phantom. The small mismatch between surface node locations on a mesh and the actual object surface arise due to measurement and manufacturing errors that cannot be completely eliminated. Additionally, geometry discretization into linear tetrahedral mesh elements introduces small errors by representing curved model surfaces with small straight lines.

A point projection algorithm allows the conversion of external phantom geometry to FE mesh coordinates. Inputs to this algorithm are node locations and element incidence lists defining both the tetrahedral mesh elements, and the triangular surface elements on the exterior surface of the mesh.

Surface projection of a reference point position involves first identifying the centroid of the reference point's motion path. The extent of the motion path is determined by enclosing the measured point position over all frames, $\bar{\mathbf{P}}$, within a rectangular volume. The side lengths of this volume, Δx , Δy and Δz , are shown in Fig. 4(c), and are used to calculate the path centroid,

$$\mathbf{P}^\Delta = \begin{pmatrix} \Delta x/2 \\ \Delta y/2 \\ \Delta z/2 \end{pmatrix}, \quad (1)$$

where $\mathbf{P}^\Delta$ is assumed to be the neutral position of the reference point motion in world space, as shown in Fig. 4(d).

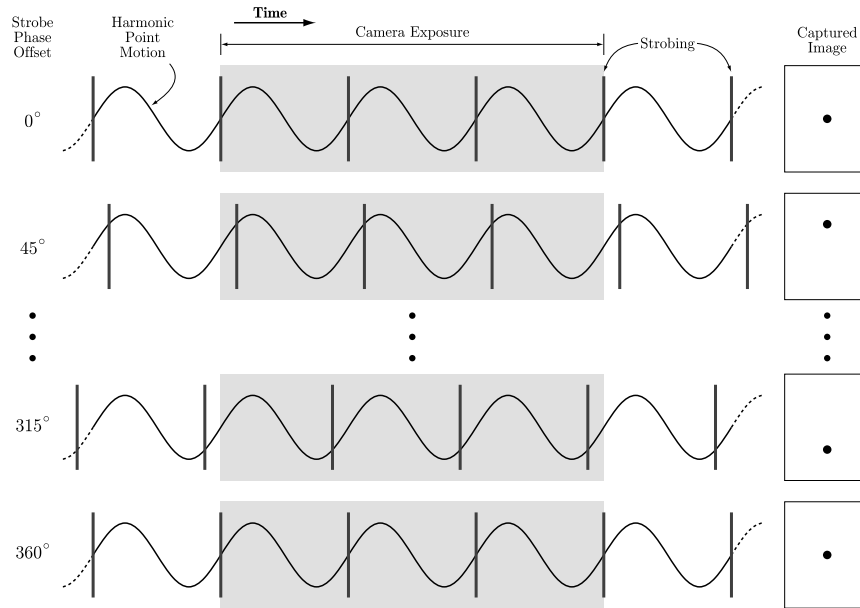


Fig. 2 – A schematic detailing the timing of the image capture process, where in this example the strobe offset between consecutive frames is 45°.

Fig. 5 is a pictorial representation of the major steps in the point projection algorithm. Surface node normal vectors shown in panel (b) are used along with linear basis function values to identify the surface element with a projection that bounds the reference point centroid. The basis function values corresponding to the projected surface element are then used to project the elevated point position to the surface of the FE mesh, as shown in panels (c) and (d).

2.3.3. Measured data fitting

The steady state oscillation of a point about an origin can be expressed using a time-harmonic displacement vector, defined at any time t as

$$\bar{\mathbf{u}}(t) = \Re\{\mathbf{u}\mathbf{e}^{i\omega t}\}, \quad (2)$$

where ω is the frequency of the system, and the amplitude term contains both real and imaginary components, $\mathbf{u} = \mathbf{u}_R + i\mathbf{u}_I$. Alternatively, reference point motion can be described using a real-valued amplitude, $\bar{\mathbf{u}}$, and a phase, φ , yielding

$$\bar{\mathbf{u}}(t) = \Re\{\bar{\mathbf{u}}\mathbf{e}^{i(\omega t + \varphi)}\}. \quad (3)$$

The measured position of a reference point at time t_i , $\bar{\mathbf{P}}^i$, can be approximated as the addition of a harmonic displacement, $\bar{\mathbf{u}}(t_i)$, to the reference point centroid, $\mathbf{P}^\Delta$, as

$$\bar{\mathbf{P}}^i = \bar{\mathbf{u}}(t_i) + \mathbf{P}^\Delta. \quad (4)$$

When assuming the measured motion track, $\bar{\mathbf{P}}$, is generated by a material that has some internal damping, (3) can be used to generate a motion fit defined using a damped amplitude, $\bar{\mathbf{u}}$, and phase, φ . A two stage algorithm is used to optimize the amplitude and phase values, where the first stage creates initial estimates, and a second stage refines the estimates using nonlinear squared error optimization. An example of the motion fit generated by this algorithm is shown in Fig. 4(d), where an ellipse has been fitted to the measured point positions described by $\bar{\mathbf{P}}$.

2.4. Stiffness parameter reconstruction

Model-based elastography differs from strain-based methods in that stress values are allowed to vary across the domain of interest. These problems require the solution of an inverse problem to determine an elastic property distribution. This inverse problem presents a significant challenge given recent results that have indicated that these forms of problems are ill-posed (Barbone and Gokhale, 2004). However, studies in both transient and steady state model-based ultrasound elastography have yielded comparable results to the strain-based approach, indicating that the mathematical challenge of non-uniqueness can be overcome with careful choice of inverse algorithm (Kallel and Bertrand, 1996; Doyley et al., 2005; Bercoff et al., 2003).

2.4.1. Finite element simulation

A FE model matching the cylindrical phantom geometry was generated and meshed using Gambit™. The surface projection technique described previously allows a single FE mesh to represent all silicone phantom models, as the three phantoms have the same external dimensions of 74 mm diameter, 74 mm height. The final model used for this experiment contained approximately 92,000 linear tetrahedral elements, with an average element edge length of 3 mm. A damped material model was used for FE simulation, where the damping ratio was fixed for the soft material as $\zeta = 0.22$, and for the hard material as $\zeta = 0.20$, based on global optimization against homogeneous material samples (Peters et al., 2008). In practice, this damping parameter can be reasonably estimated by fitting a homogeneous assumption to the data, thus allowing the damping ratio to be considered as *a priori* information during the reconstruction process.

To match the experimental setup, the FE model was displacement constrained in all three axes on the lower face, with x and y displacements set to zero, and sinusoidal

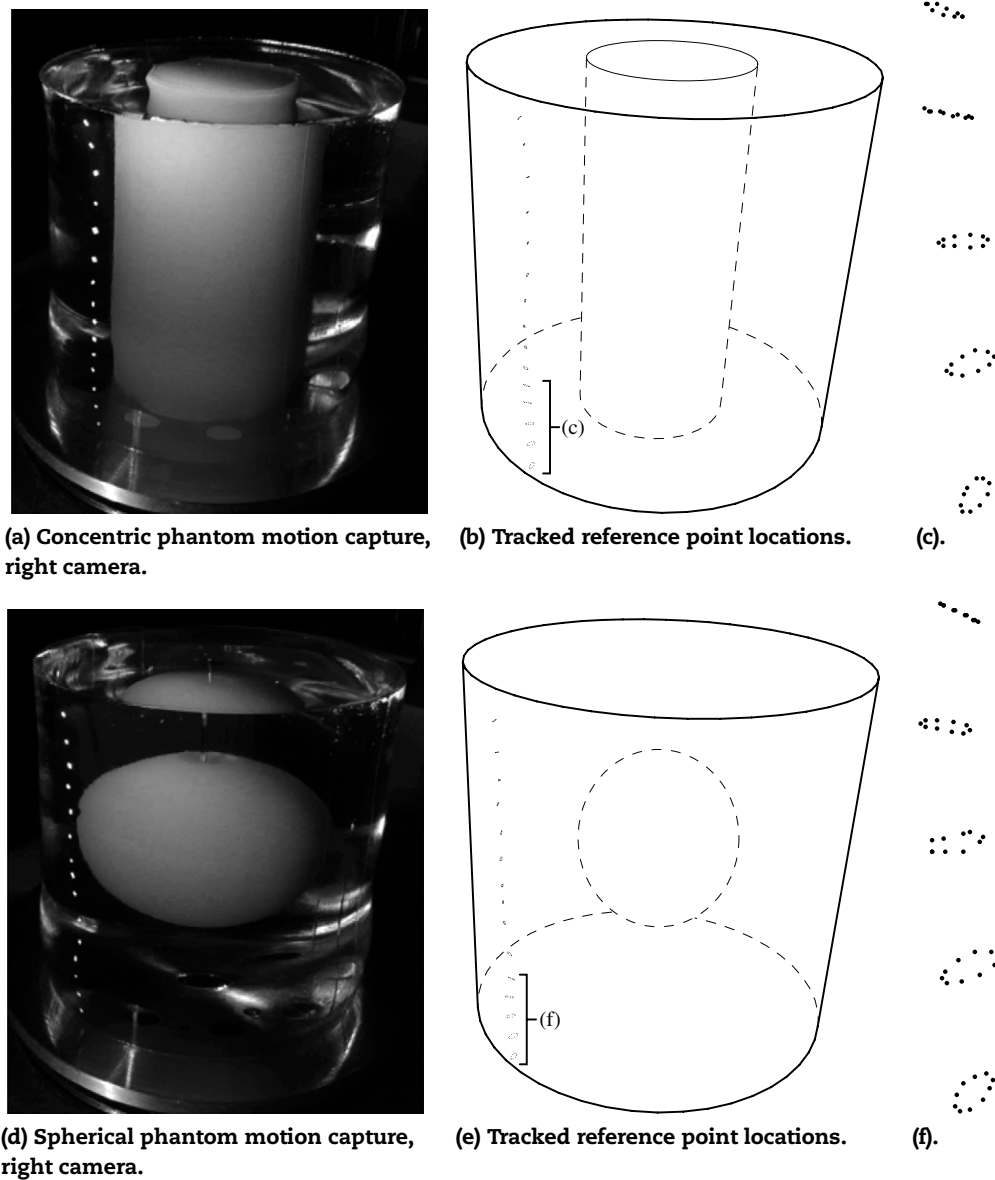


Fig. 3 – Experimentally observed motion of the two heterogeneous silicone phantoms (NOTE: for clarity, only 10 of the 20 captured motion frames are shown in this figure).

actuation applied in the z direction at 0.5 mm amplitude, 100 Hz. A constant density value of 1000 kg/m^3 was used within the model. FE calculations were made using the author's own software, which is integrated directly into the reconstruction scheme.

2.4.2. Shape-based property parameterization

To reduce the number of parameters required for elastic property reconstruction in the phantoms discussed here, a shape-based material property parameterization was implemented to define the stiffness distributions used for the FE calculations. For the concentric phantom, these shape parameters were defined as E'_s , representing the modulus of the soft, exterior silicone, E'_h , representing the modulus of the hard, silicone inclusion, and P_c , representing the interface position, defined as the ratio of the inclusion diameter to the total phantom diameter. For the spherical

and homogeneous soft phantoms, the shape parameters were defined as E'_s , again the modulus of the soft silicone exterior, P_d , representing the ratio of the inclusion diameter to the total phantom diameter, and P_h , representing the height of the inclusion from base of the phantom. For these two phantoms, the stiff inclusion modulus, E'_h , was assumed to be 103 kPa based on global optimization against homogeneous material samples (Peters et al., 2008).

2.4.3. Surface error objective function

Once the experimental motion data is represented in the same coordinate space as the FE model, a quantitative match between the simulated and measured motions can be evaluated. The damped FE model provides calculated displacement amplitudes containing both real and imaginary components, $\mathbf{u}^c = \mathbf{u}_R^c + i\mathbf{u}_I^c$. The measured surface motion is fitted using damped amplitude, $\tilde{\mathbf{u}}$, and phase, φ , and is

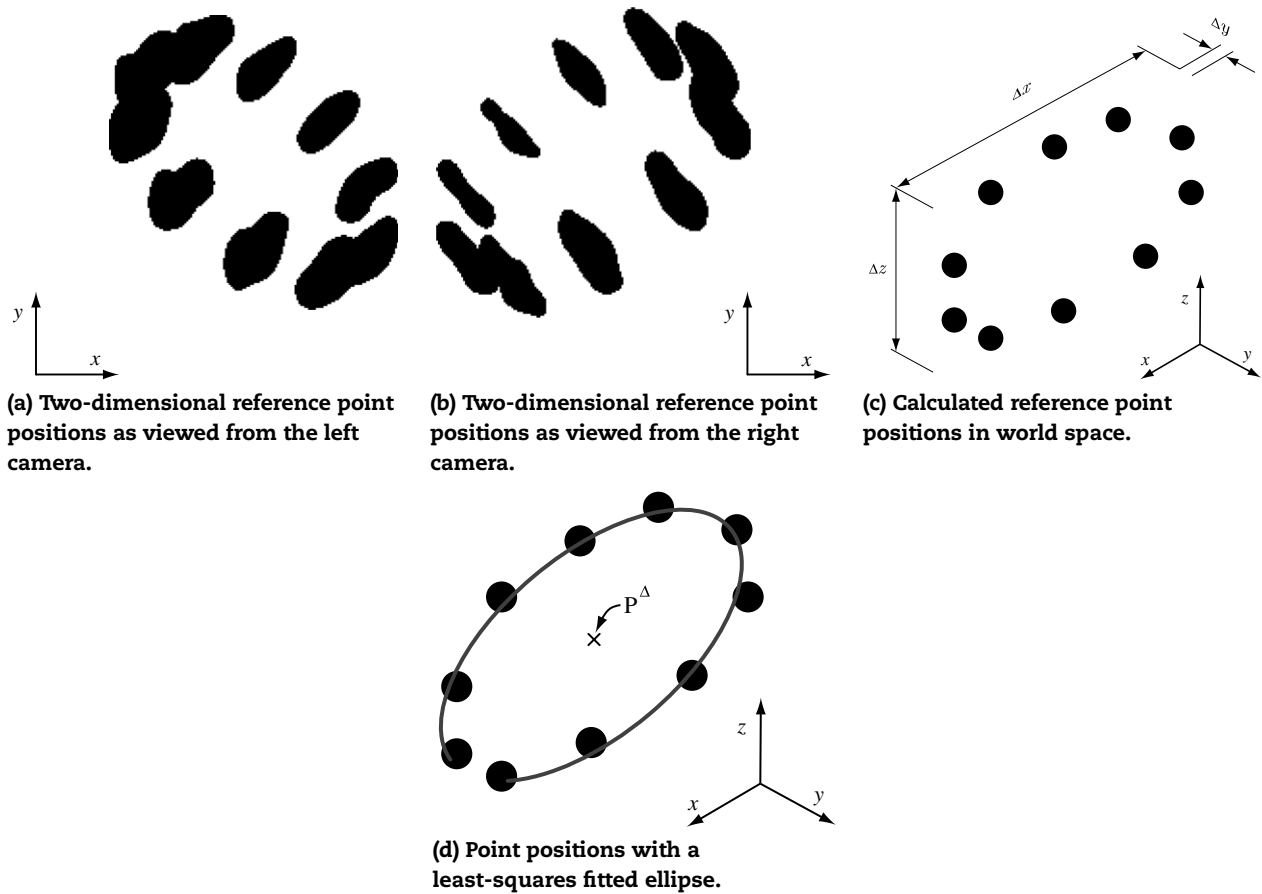


Fig. 4 – Conversion and fitting of reference point position from two-dimensional to three-dimensional coordinate systems (NOTE: for clarity, only 10 of the 20 captured motion frames are shown in this figure).

expressed as real and imaginary components, $\mathbf{u}^m = \mathbf{u}_R^m + i\mathbf{u}_I^m$. Here, the simulated displacement amplitude, $\mathbf{u}^c$, can be compared to the measured displacement amplitude, $\mathbf{u}^m$, at all N reference points, using

$$\Omega = \frac{1}{3N} \sum_{i=1}^N (|\mathbf{u}_R^m - \mathbf{u}_R^c| + |\mathbf{u}_I^m - \mathbf{u}_I^c|)_i, \quad (5)$$

where Ω forms the objective function for all reconstruction efforts.

2.4.4. Hybrid reconstruction algorithm

The hybrid optimization algorithm used in this study is a combination of traditional Gradient Descent (GD) and novel Combinatorial Optimization (CO) methods. The CO phase of the algorithm combines features of two popular CO techniques — Genetic Algorithms (GA) and Simulated Annealing (SA) (Holland, 1992; Carnahan and Sinha, 2001; Peters et al., 2008). The CO algorithm contains parameter pairs, G_i^j , defined as *genes*, where the subscript refers to the i th gene in the population, $1 \leq i \leq N_G$, and the superscript, j , indicates the gene's *generation*, where $0 \leq j \leq M$. Objective function values, Ω_i^j , are defined using the same convention. Values for both N_G and M can be chosen arbitrarily, with an upper limit usually reflecting increased computational cost. Within an entire population of genes at a particular generation, termed G^j , the gene providing the

closest match to experimental data is described as G_+^j , and has corresponding error metric value, Ω_+^j .

CO begins with a population of N_G genes, each containing an initial value for the two parameters, $G_i^0 = \{P, C\}_i^0$. Initial values for all parameters within all genes are usually randomly assigned within a range based on *a priori* knowledge. Before entering the CO algorithm proper, FE simulations are performed to obtain starting values of the error metric, Ω_i^0 , for all genes in the population.

Mating in CO involves interchanging genetic material between genes, with the intent of improving the fitness of the population as a whole. Pairs of genes are identified for mating using roulette wheel selection, where the probability of gene selection is directly proportional to the value of its error metric, Ω_i^j (Zhong et al., 2005). Genes selected as one of a mating pair have their constituent parameter values modified by swapping a proportion of their binary expression with bits from the corresponding parameter in their mating partner. The chance of the mating operation being performed is set with the probability value Π_{mate} . Parameter values output from the mating process form a new, temporary gene pool.

Random mutation modifies the value of any parameter within any gene by flipping the value of a bit from the parameter's binary expression. This process increases the diversity of the population by introducing new genetic material that is unlikely to be introduced through mating

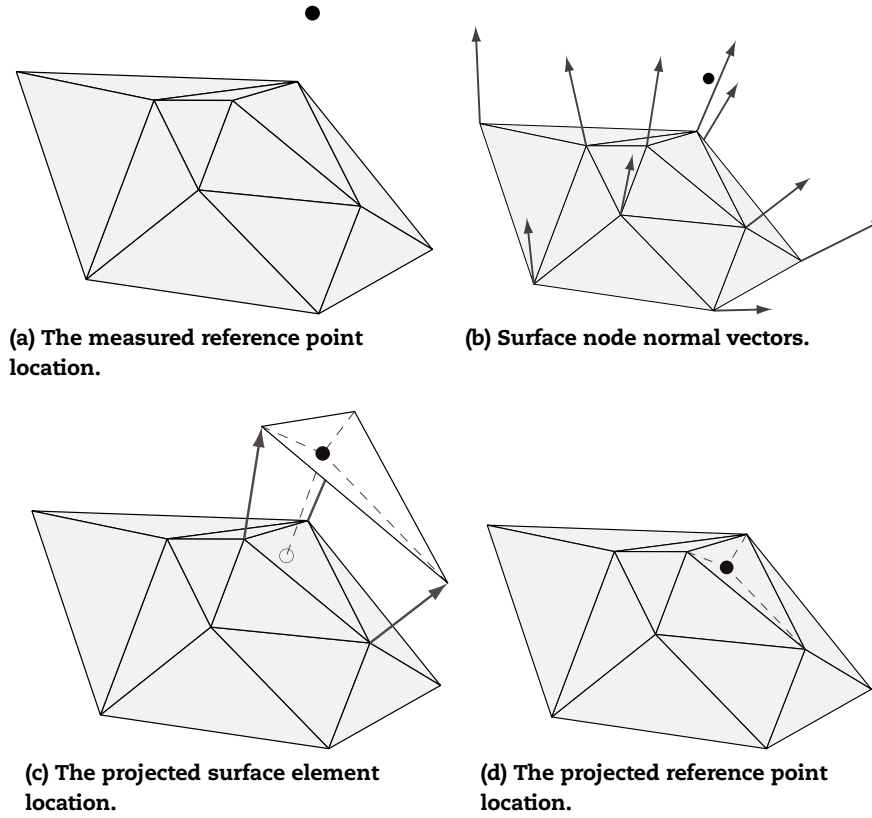


Fig. 5 – A graphical representation of the reference point projection process to the surface mesh.

alone. Mutation is potentially performed on every bit within the temporary population after mating, where the likelihood of a bit changing value is governed by the probability Π_{mutate} . The resulting population is defined as $\tilde{G}_i^j$, where the tilde indicates that the mating and mutation operations have taken place. FE simulation at the parameter values provided by these potential genes allows corresponding error metric values, $\tilde{\Omega}_i^j$, to be calculated.

SA takes its theoretical approach from the statistical mechanics of physical annealing, where solutions that decrease the global system ‘energy’, here the error term Ω , are increasingly preferred as a system-wide temperature variable is decreased over time (Kirkpatrick et al., 1983). In the CO algorithm described here, SA provides a mechanism for determining which genes from the potential offspring population, $\tilde{G}^j$, will become part of the subsequent generation, G^{j+1} . Error values from each potential gene, $\tilde{\Omega}_i^j$, are compared with the corresponding value from the current population, Ω_i^j . The chance of the offspring gene, $\tilde{G}_i^j$, replacing the original gene, G_i^j , in the subsequent population is set by the probability $SA\Pi_i^j$, defined

$$SA\Pi_i^j = \exp\left(\frac{-\tilde{\Omega}_i^j/\Omega_i^j}{T^j}\right), \quad (6)$$

where T^j is the value of the system-wide temperature variable at generation j . Decreasing T each generation means potential genes that would increase the population error

have a decreasing chance of being selected as generations progress. In this experiment, the temperature was decreased logarithmically between values of 10 and 0.5 over a total of 100 generations. After completing SA screening, parameter values from all genes chosen for the subsequent generation are logged, along with their error values, and the gene displaying the best match to the experimental data, G_+^j , noted.

The CO algorithm in this study had two stopping criteria:

1. The error metric from the best-fitting member of the current population, Ω_+^j , was less than a pre-determined minimum based on the geometry.
2. The maximum number of generations for the reconstruction, M , was reached.

Upon halting the CO algorithm, the parameter values providing the closest match to the measured motion, $G_+ = (P, C)_+$, with corresponding error metric, Ω_+ , were obtained.

3. Results

The experiments performed in this study first aimed to optimize the performance of the hybrid algorithm as applied to the concentric silicone phantom data, before applying the algorithm to the spherical phantom reconstruction. Performing a sensitivity analysis on the hybrid algorithm allows the influence of different algorithm settings on the reconstruction behavior to be assessed, with regards both to solution accuracy and computational efficiency.

3.1. Hybrid algorithm sensitivity analysis

The sensitivity analysis considered independent variation in the three parameters for the concentric phantom: E'_s , E'_h and P_c . The use of three parameters provides a large and complicated error domain likely to contain many local minima. All reconstructions performed during this sensitivity study used the experimentally measured concentric motion data as the reference displacement set, and calculated the surface error metric Ω according to (5).

Hybrid reconstruction algorithm optimization was performed by varying several key settings controlling both CO and GD stages. As both phases are performed separately in the hybrid method, this optimization was performed on each component individually, before combining both parts. As this form of reconstruction algorithm is highly application specific, the optimized algorithm settings chosen represent the ideal case for the specific shape-based parameter reconstruction considered. To avoid initial bias in the CO sensitivity study, all reconstructions start from the same initial population set, G^0 .

3.1.1. Combinatorial optimization algorithm

The investigation of algorithm settings for the CO optimization involved the results of 100 reconstructions, each based on a different initial population, $^lG^0$, and generating output $^lG_+$, where $1 \leq l \leq 100$. These results were normalized to give a measure of reconstruction performance between 1.0 and 0.0, where 1.0 represented the mean error at generation zero,

$$\bar{\Omega}_+^0 = \frac{1}{100} \sum_{l=1}^{100} ({}^l\Omega_+^0), \quad (7)$$

and 0.0 indicated the minimum error value within the domain, $\hat{\Omega}$. To allow performance characterization, the best-fit parameter values leading to $\hat{\Omega}$, termed $\{\hat{A}, \hat{B}, \hat{C}\}$, were determined prior to the study.

The first phase of CO optimization varied the value of Π_{mate} from 0.2 to 0.8 in increments of 0.2. For these reconstructions, Π_{mutate} was set to zero to eliminate the influence of random mutation on the reconstruction results, and the number of genes was fixed, $N_G = 10$. The resulting CO algorithm performance across 100 generations is shown in Fig. 6(a), where each line represents the mean of 100 individual reconstructions. Optimal reconstruction performance was seen when $\Pi_{mate} = 0.6$, where the normalized algorithm performance value was 0.18 after 100 generations.

CO sensitivity to mutation was tested by fixing Π_{mate} at the optimized value of 0.6, and varying Π_{mutate} from 0.0 to 0.2. The resulting normalized CO algorithm performance across 100 generations is shown in Fig. 6(b). In this plot, the solid line corresponds to the $\Pi_{mate} = 0.6$ result in Fig. 6(a). No significant improvement in performance was noted at Π_{mutate} values greater than 0.1, where the normalized performance value was 0.05 after 100 generations.

Finally, the number of genes in the population, N_G , was varied, with $\Pi_{mate} = 0.6$, and $\Pi_{mutate} = 0.1$. CO reconstructions containing 5, 10, 20 and 40 genes were performed, with results shown in Fig. 6(c). In this plot the

Table 1 – Optimal settings for the CO algorithm determined via sensitivity analysis.

Setting	Value
Π_{mate}	0.6
Π_{mutate}	0.1
N_G	10

starting values for the performance metric varied in each case, as the different values for N_G led to a range of values for $\bar{\Omega}_+^0$. To normalize for the increased computational cost of updating additional genes through each generation, the horizontal axis of this figure indicates the total number of FE simulations performed. Regardless of this additional normalization, adding genes to the population continued to improve algorithm performance. A value of $N_G = 10$ was selected as being optimal for this problem, given the physical number of computer workstations available, and the absence of significant performance improvement between 10 and 20 genes. The optimized CO algorithm settings used for subsequent parameter reconstruction are summarized in Table 1.

3.1.2. Gradient descent algorithm

The simple three-parameter GD algorithm used in this study did not require Hessian matrix regularization for numerical stability. This inherent stability limited the GD algorithm optimization study to the total number of iterations performed. Fig. 6(d) shows the normalized performance result from 100 GD reconstructions starting at random parameter values within the domain. The performance metric from all reconstructions was tracked over the total of 20 iterations performed, with no improvement in performance observed beyond 10 iterations. The absence of further improvement in the GD results beyond 10 iterations indicated that all instances of the algorithm had converged prior to this point, where the normalized performance metric was 0.66. The high minimum value of the performance metric in Fig. 6(d) indicates that the majority of GD reconstructions converged to a parameter combination representing local error minima that were far from the globally optimum solution.

3.2. Concentric phantom reconstructions

After optimizing the performance of the CO and GD components of the hybrid algorithm, reconstructions were performed on the experimental phantom data using GD and CO separately, followed by the hybrid algorithm that combined both techniques. Each trial reconstruction method was implemented 100 times to provide statistical evidence of the best solution. The mean final reconstructed value for each parameter over all 100 reconstructions, $\bar{A}_+$, $\bar{B}_+$, $\bar{C}_+$, was compared to the best-fit value for that parameter, $\hat{A}$, $\hat{B}$, $\hat{C}$, using two metrics. The ψ metric acts as a measure of the mean normalized parameter error,

$$\psi = \frac{|\hat{A} - \bar{A}_+|}{|\hat{A}|} + \frac{|\hat{B} - \bar{B}_+|}{|\hat{B}|} + \frac{|\hat{C} - \bar{C}_+|}{|\hat{C}|}. \quad (8)$$

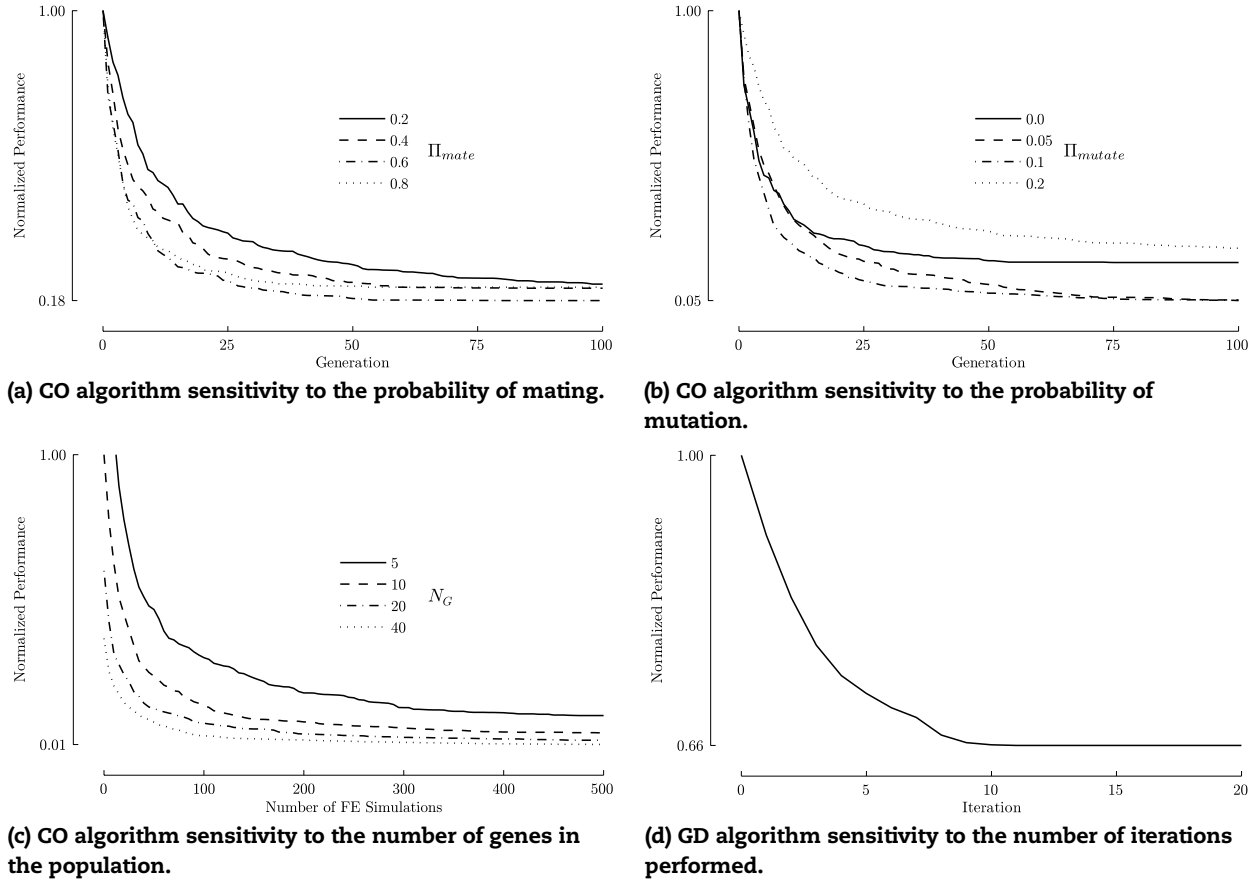


Fig. 6 – Results from the sensitivity analysis performed on the CO and GD reconstruction algorithms.

Additionally, Σ^2 was used as a measure of the normalized variance of the reconstructed solutions, defined as

$$\Sigma^2 = \sum_{l=1}^{N=100} \frac{(\hat{A}_l - \hat{A})^2}{(N-1)\hat{A}^2} + \frac{(\hat{B}_l - \hat{B})^2}{(N-1)\hat{B}^2} + \frac{(\hat{C}_l - \hat{C})^2}{(N-1)\hat{C}^2}. \quad (9)$$

In the following results, the example parameter values $\{A, B, C\}$ are represented by the shape-based parameters relevant to this experiment, $\{E'_s, E'_h, P_C\}$.

Fig. 7 is the resulting stiffness solution distribution after performing 100 GD reconstructions on data from the concentric phantom. Each reconstruction had random starting values and was run for a total of 10 iterations. Reconstruction error metrics Ψ and Σ^2 were 0.79 and 0.55 respectively. The finite-difference method used for the Jacobian matrix calculation meant that a total of 40 FE simulations were required for each reconstruction. The best-fit stiffness profile on the plot represents the stiffness distribution from the parameter values corresponding to $\hat{\Omega}$, while the measured values correspond to data from mechanical testing and size measurements.

Fig. 8 are the result of the optimized CO algorithm applied to the concentric phantom displacements, where results at a range of generations are presented. Fig. 8(a)–(f) shows results from the hybrid algorithm applied to the same data, where optimized GD reconstruction was performed following CO, using a starting guess for the parameters provided by the best result from the CO reconstructions, G_+ . Table 2

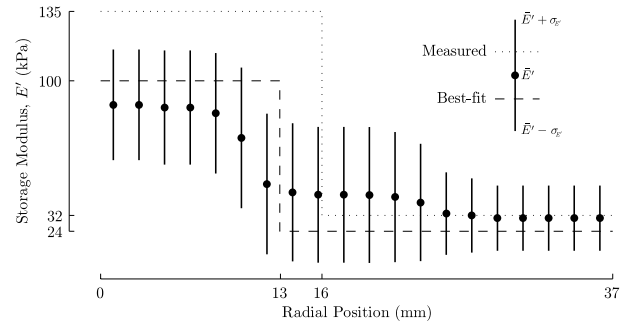


Fig. 7 – The reconstructed stiffness profile distribution from 100 GD reconstructions on concentric phantom data, where the measured stiffness profile, and best-fit profile based on $\hat{\Omega}$, are indicated.

summarizes the performance metric results across the optimized reconstruction algorithms considered, where the values of Ψ and Σ^2 for the hybrid algorithm are consistently less than the corresponding CO or GD algorithms.

3.3. Spherical phantom reconstructions

The elastic reconstruction of the spherical phantom considered a total of three parameters, allowing the hybrid algorithm applied to the concentric phantom motion to be

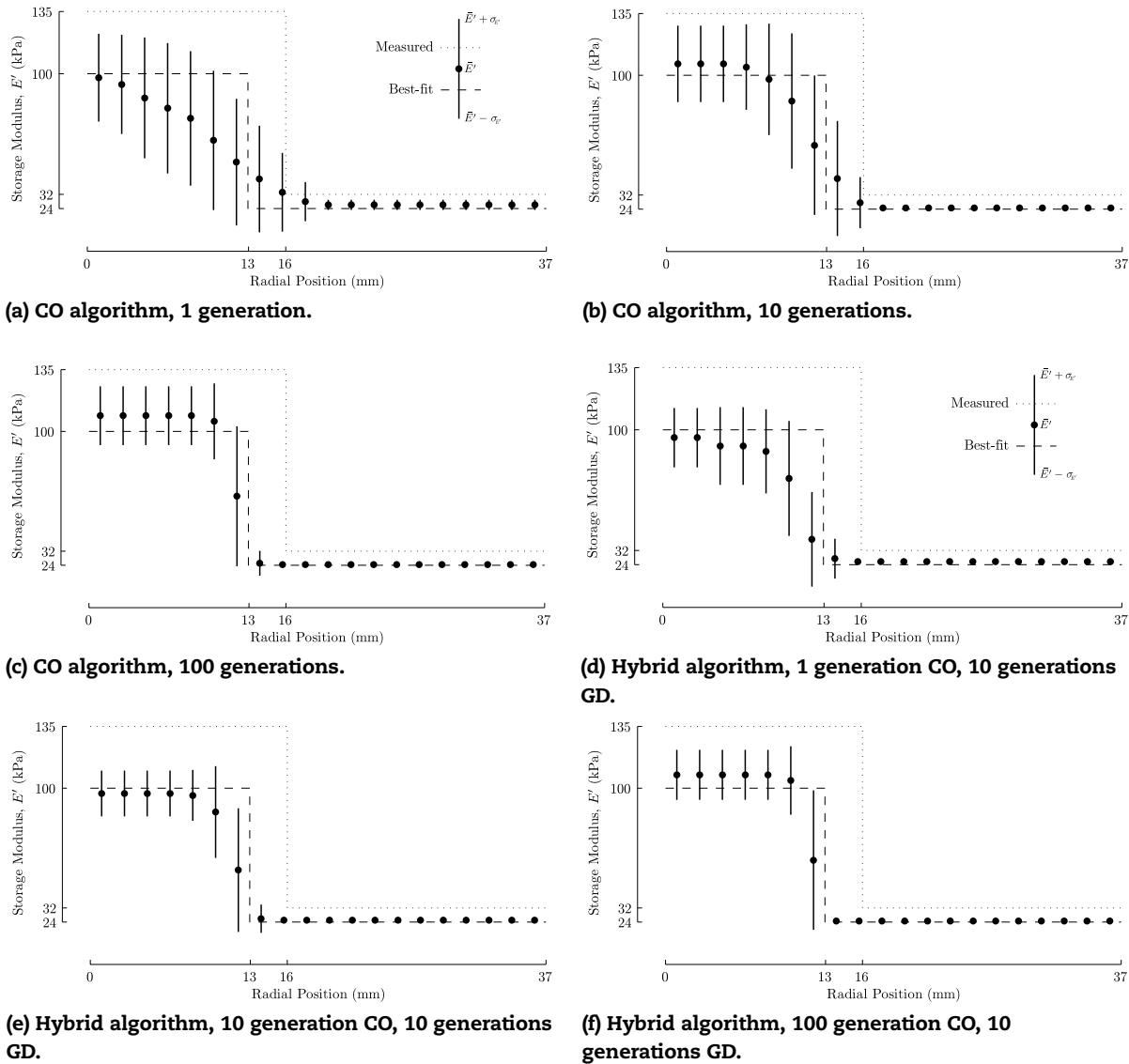


Fig. 8 – The stiffness profile distribution from 100 CO and hybrid reconstructions on concentric phantom data, where the measured stiffness profile, and best-fit profile based on $\hat{\Omega}$, are indicated.

Table 2 – Results from the hybrid algorithm optimization.

Algorithm	Iterations		Metric		Consecutive solves	
	CO	GD	Ψ	Σ^2	Serial (actual)	Parallel (theory)
GD	–	10	0.79	0.55	40	10
CO	1	–	0.64	0.19	10	1
	10	–	0.41	0.09	100	10
	100	–	0.25	0.04	1000	100
Hybrid	1	10	0.39	0.07	50	11
	10	10	0.25	0.03	140	20
	100	10	0.23	0.03	1040	110

reused with minimal modification. A *contradictive* approach to internal stiffness estimation requires any hard inclusion to have a constant stiffness value, meaning the value of E'_h was fixed, and hence no longer a reconstructed parameter

(Peters et al., 2009). Removing this second stiffness parameter allows the more complicated spherical phantom geometry to be described using an additional positional value without increasing the total number of independent parameters. Here, the ratio of the spherical inclusion diameter to the overall phantom diameter is described by the parameter P_d . Physical measurements of the phantom give an actual value for this parameter of $\hat{P}_d = 57\%$. The vertical position of the spherical inclusion, which was assumed to always lie along the central axis of the cylinder, is described by the height parameter P_h , where $\hat{P}_h = 50\%$.

A first attempt at three-parameter reconstruction of the spherical phantom was made using the hybrid algorithm settings in Table 2, where $I_{mate} = 0.6$, $I_{mutate} = 0.2$, and the number of genes was set at $N_G = 10$. Both CO and GD phases of the hybrid algorithm were run for 10 iterations. Fig. 9(a) shows the mean stiffness distribution across 100 hybrid reconstructions performed on the spherical geometry, each started from a different initial guess distributed across

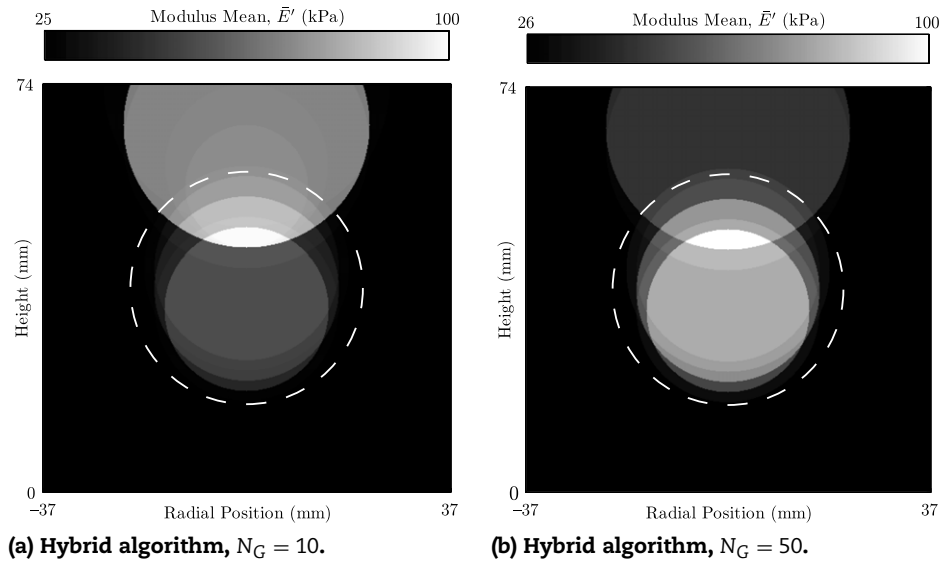


Fig. 9 – A spatial representation of the spherical phantom modulus reconstructed using the hybrid algorithm with varying numbers of genes.

the parameter domain. The plane of the figure passes through the center of the cylindrical phantom, and the stiffness value at each point within the image is calculated as the mean indicated stiffness value over all reconstruction results. The measured position and size of the actual phantom inclusion is indicated as a dashed circle. Two distinct regions of high stiffness are visible, where an area of high stiffness near the top of the phantom is indicated in addition to the expected centrally located hard region.

The inconclusive result in Fig. 9(a) led to an adjustment in the hybrid algorithm settings. The sensitivity analysis study found that increasing the number of genes in the CO phase of the hybrid algorithm had a significant performance benefit, even when considering the increased computational cost associated with the larger population. With this result in mind, reconstructions were performed on the spherical geometry using greater values for N_G , where the stiffness distribution resulting from hybrid reconstructions performed with $N_G = 50$ is shown in Fig. 9(b). Here, improvement can be seen in the spatial modulus distribution, where the presence of the large artifact near the top of the phantom is somewhat diminished.

Further testing led to satisfactory reconstruction performance with a value of $N_G = 100$. Fig. 10(a) shows the resulting spatial modulus distribution from an average of 100 different hybrid reconstructions at this value. In this case, the result strongly indicated the presence of a central inclusion location at the measured height, with a slightly smaller diameter. The reconstructed soft modulus value was approximately $E'_s = 26$ kPa. The variance across the 100 different reconstructed modulus distributions within the phantom is shown in Fig. 10(b), which indicates that the highest variance values occur near the edges of the central inclusion. Despite the relatively conclusive stiffness reconstruction in this phantom, there is an observable artifact toward the top of this phantom stiffness image, Fig. 10(a), although this is seen to correspond to a region of increased of uncertainty in Fig. 10(b).

3.4. Homogeneous soft phantom reconstructions

The specificity of the hybrid algorithm applied to the spherical geometry was evaluated by reconstructing internal stiffness using the spherical model and the experimental motion data obtained from the soft homogeneous silicone phantom. In this case, the expected result was an entirely soft distribution free from significant hard inclusions. The reconstructed stiffness distribution obtained is shown in Fig. 10(c)–(d), where all reconstructions used the same algorithm settings as those used when performing the reconstructions against the spherical phantom data with $N_G = 100$. Two small areas of moderate stiffness are visible in the resulting plots at positions near the top of the model. In both cases, the modulus variance in the vicinity of the areas of increased stiffness is correspondingly high.

4. Discussion

The results shown in Figs. 7–10 are the distribution of the spatially interpreted stiffness solutions resulting from performing 100 of the respective reconstructions. As such, they should not be interpreted as the ‘result’ of a reconstruction, but rather a distribution drawn from several reconstruction results. For the concentric reconstructions, reduced values of Ψ and Σ^2 are desirable, as they indicate both reduced mean error and reduced uncertainty, which is important when performing a limited number of reconstructions on a real data set.

The distribution of GD reconstruction results shown in Fig. 7 do not give a close match to the best-fit phantom stiffness distribution, with reconstruction metrics of $\Psi = 0.79$, $\Sigma^2 = 0.55$. The large variance visible in the plot at all radial positions indicates that the GD reconstructions failed to converge consistently for the parameters E'_s , E'_h and P_C . Even using a damped material model, it is apparent that local

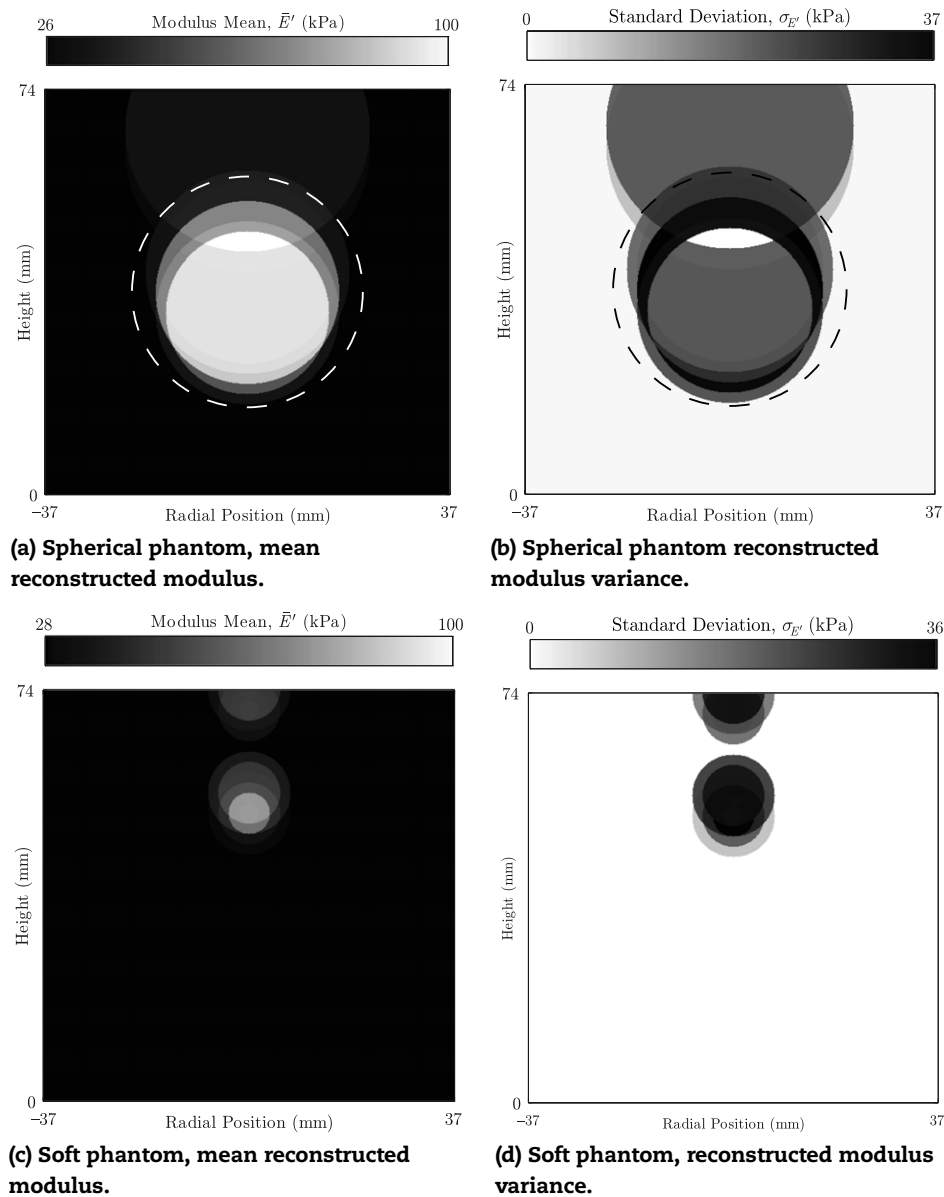


Fig. 10 – Silicone phantom modulus distributions reconstructed using a contradicitive hybrid algorithm with 100 genes.

minima severely hamper the algorithm performance, a result consistent with previous studies applying the GD method to reconstructions with fewer parameters (Peters et al., 2006).

The accuracy of reconstructed solutions showed a marked improvement when using an optimized CO approach. Mean error and variance in the stiffness profile distributions in Fig. 8(a)–(c) are significantly lower than the GD case, particularly in the estimation of the soft modulus value, E'_s . Improvement is also visible in the definition of the step stiffness change between hard and soft regions. Performance metrics of $\Psi = 0.25$ and $\Sigma^2 = 0.04$ were achieved after 100 generations, where a total of 1000 FE simulations were required for each reconstruction.

Adding GD reconstruction to CO to form a hybrid method led to further improvements in the distribution of results in all cases shown in Fig. 8(d)–(f). Notably, the total number

of FE solves required to obtain the hybrid result in Fig. 8(e) is only 14% of the number of solves required to achieve the CO result shown in Fig. 8(c). This result demonstrates that a hybrid approach can achieve equivalent performance to CO performed alone, while requiring significantly fewer computational resources.

Table 2 includes the number of FE simulations required for each instance of the three algorithms considered. The results from this study using a serial computation approach show that performing fewer than 100 CO generations can significantly reduce computational cost, while retaining a significant reconstruction performance advantage in relation to the GD algorithm. When considering a parallel implementation of the hybrid algorithm, where independent FE solves are performed concurrently across multiple processors, the total number of consecutive FE solves reduces

to 20 for an algorithm comprising 10 generations of CO and 10 iterations of GD reconstruction.

The reconstructed results in Fig. 9 indicate that the three-dimensional error domain for the spherical reconstruction problem does not contain an obvious single minimum value. A bi-modal distribution is apparent, with Fig. 9(a) indicating that approximately half of all reconstructions predicted a large inclusion located near the top of the phantom model that would be visible on the upper phantom surface. While the use of CO in previous reconstructions allowed the hybrid algorithm to avoid such local minima, the presence of two potential solutions with very similar surface error metric values presents a significant challenge to the hybrid algorithm when using a population comprising 10 genes.

The improvement in reconstructed modulus distribution when adding genes to the CO phase of the hybrid reconstruction indicates that while the presence of the upper inclusion was the result of a local error minimum, the global error minimum $\hat{\Omega}$ closely corresponds to parameters describing the actual phantom geometry. Additional genes increase the diversity of the CO population, meaning that mating and mutation have a greater chance of creating an offspring gene closer to the globally optimal parameter combination. The population diversity when $N_G = 100$ reached a level at which the globally optimum solution was preferred by almost all reconstructions.

The existence of substantial local minima in the spherical case is likely due to a lack of motion data from the upper surface of the phantom. Previous surface error studies performed on a ‘stacked’ phantom geometry indicate that when surface reference points are located across an external boundary between soft and hard material, the existence and position of such an interface can be clearly determined (Peters et al., 2009). Additional reference points on the top surface of the phantom should allow the motion error metric to clearly identify the presence or absence of inclusions near or at the top surface of the model based on their surface displacement values.

The slight underestimation of the size of the inclusions in Figs. 8 and 10 is likely a result of the nodal stiffness interpolation used in the FE model. Additionally, the rounding of P_d during the spherical reconstructions may have led to a small underestimation of the predicted inclusion diameter. The effects of this property interpolation should be reduced with finer nodal spacing in the FE mesh used for simulations.

The specificity analysis of the hybrid reconstruction algorithm using the homogeneous phantom experimental data indicated the presence of small, moderate stiffness regions near the top of the phantom. While the soft modulus value reconstructed across the majority of the phantom was in good agreement with the expected result, the small areas of elevated stiffness are undesirable, as the ability of any screening system to minimize the occurrence of false positive tests is important. This result is further evidence that motion data at the upper phantom surface may improve the ability of the surface error metric to correctly eliminate geometries that predict the presence of inclusions near the top of the model.

Despite the challenges presented by both error domains, the presence of a hard inclusion in the spherical case was

Table 3 – Computational cost considerations for the hybrid algorithm applied to the spherical silicone phantom.

Algorithm	N_G	Consecutive solves	
		Serial (actual)	Parallel (theory)
Hybrid	10	140	20
	50	540	20
	100	1040	20

more strongly suggested than the presence of any hard area in the homogeneous phantom. All areas of elevated stiffness corresponded to the highest observed values for modulus variance in the homogeneous phantom reconstructions. This result differs from the spherical case, where variance values within the interior of the hard inclusion were lower, and the highest variance values were around the edge of the inclusion, as expected.

Increasing the number of genes in the CO stage of the hybrid reconstruction gives a subsequent increase in computational cost, summarized in Table 3. Here, the hybrid reconstructions with $N_G = 100$ require 10 times the number of FE solves as the baseline case where $N_G = 10$, when performed on a serial system. However, the inherent suitability of CO to parallel computation means that if performed on a large parallel system of processors, an increase in genes would not increase the overall time required for computation, as each gene would be simulated on a dedicated processor. Additionally, it should be noted that all the hybrid reconstructions required significantly less computational effort than an exhaustive search within the problem domain. For the spherical geometry considered in this case, with a total of three independent parameters, an exhaustive search would involve a total of 4851 FE simulations.

In general, the parameterized, shape-based property descriptions used in this study act as a significant regularization method for this ill-posed inverse problem. Obviously, in the context of breast cancer screening, the axisymmetric constraints on the property descriptions used here will have to be lifted, and the assumption of homogeneous, soft background material may have to be modified as well. To maintain an ability to obtain property estimates with some level of statistical certainty, the effects of these additional degrees of freedom on the condition of the inverse problem will have to be ameliorated by improvements in the quality of the data, most likely in the form of more accurate and more complete surface motion descriptions. These improvements will require substantial development from the prototype imaging system used to obtain the results described here, and thus this effort must be considered as *future work* until a more sophisticated imaging system is assembled.

5. Conclusions

The hybrid algorithm optimization study described in this section demonstrated the ability of a three-parameter approach to reconstruct heterogeneous phantom stiffness us-

ing experimentally measured displacements. Though the computational cost was high, the CO method obtained significantly better reconstruction results than a GD approach, confirming the findings of the previous simulation study. The optimized hybrid reconstruction method demonstrated an ability to better the performance of CO performed alone, with almost an order of magnitude reduction in computational cost. When the promising results from the contradictory approach to parameter estimation are considered, a hybrid approach to inclusion imaging has the potential to solve problems with increased complexity. Here, more realistic internal geometries can be described without increasing the total number of parameters required, by fixing the modulus value of any predicted hard inclusion. Additionally, recent developments in the use of more sophisticated, shape-based functional property descriptors indicate that much more complex internal property distributions can be considered while keeping the number reconstruction parameters for the inverse problem relatively low (Naik et al., 2009).

The final reconstruction study indicated that the hybrid reconstruction method had the ability to reconstruct the internal stiffness distribution of a more realistic geometry containing a spherical inclusion. While the algorithm required additional tuning to achieve a positive result in this case, the additional computational cost could be offset with the use of parallel computation. The continued presence of local error minima in this study highlighted the requirement of a DIET system to have accurate motion data from as much of the exposed object surface as possible to ensure that reconstruction results return the expected stiffness distribution.

6. Disclosure

The three authors of this paper are shareholders in the limited liability company *Boundary LifeSciences Ltd.*, which has commercial interests in the DIET method.

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